

$$\begin{aligned}
& y_e^{i1i2} \rightarrow \\
& -y_e^{i1i2} + \frac{1}{16\pi^2} \left(\frac{1}{16} \left(-6 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} - y_e^{i1i2} (13 g_1^2 + 3 g_2^2) \right) + g_1^2 y_e^{i1i2} \text{LF}_{1,1,0} [m_1, m_e^{i2}] - \right. \\
& \frac{1}{2} g_1^2 y_e^{i1i2} \text{LF}_{2,1,-1} [m_1, m_e^{i2}] + \frac{1}{4} g_1^2 y_e^{i1i2} \text{LF}_{1,1,0} [m_1, m_l^{i1}] - \\
& \frac{1}{8} g_1^2 y_e^{i1i2} \text{LF}_{2,1,-1} [m_1, m_l^{i1}] + \frac{3}{4} g_2^2 y_e^{i1i2} \text{LF}_{1,1,0} [m_2, m_l^{i1}] - \\
& \frac{3}{8} g_2^2 y_e^{i1i2} \text{LF}_{2,1,-1} [m_2, m_l^{i1}] + \frac{1}{2} \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{1,1,0} [m_e^r, \tilde{\mu}] - \\
& \frac{1}{2} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{2,1,0} [m_l^p, m_e^r] + \frac{1}{2} y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} \text{LF}_{3,1,-1} [m_l^p, m_e^r] - \\
& \frac{1}{2} y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi2^2} a_e^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr}) \right) \text{LF}_{3,1,0} [m_l^p, m_e^r] + \\
& \frac{3}{2} y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi2^2} a_e^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr}) \right) \text{LF}_{4,1,-1} [m_l^p, m_e^r] - \\
& y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} + \overline{a_e^{pr}} (2 C_{\phi2^2} a_e^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{pr}) \right) \text{LF}_{5,1,-2} [m_l^p, m_e^r] + \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \\
& \text{LF}_{1,1,0} [m_l^p, \tilde{\mu}] - \frac{3}{2} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{2,1,0} [m_q^p, m_d^r] + \frac{3}{2} y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} \text{LF}_{3,1,-1} [m_q^p, m_d^r] - \\
& \frac{3}{2} y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi2^2} a_d^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr}) \right) \text{LF}_{3,1,0} [m_q^p, m_d^r] + \\
& \frac{9}{2} y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi2^2} a_d^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr}) \right) \text{LF}_{4,1,-1} [m_q^p, m_d^r] - \\
& 3 y_e^{i1i2} \left(C_{\phi2\phi1} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} + \overline{a_d^{pr}} (2 C_{\phi2^2} a_d^{pr} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_d^{pr}) \right) \text{LF}_{5,1,-2} [m_q^p, m_d^r] - \\
& \frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{2,1,0} [m_u^r, m_q^p] + \frac{3}{2} \tilde{\mu}^2 y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \text{LF}_{3,1,-1} [m_u^r, m_q^p] - \\
& \frac{3}{2} \tilde{\mu} y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi2\phi1} a_u^{pr} + 2 C_{\phi2^2} \tilde{\mu} y_u^{pr}) \right) \text{LF}_{3,1,0} [m_u^r, m_q^p] + \\
& \frac{9}{2} \tilde{\mu} y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi2\phi1} a_u^{pr} + 2 C_{\phi2^2} \tilde{\mu} y_u^{pr}) \right) \text{LF}_{4,1,-1} [m_u^r, m_q^p] - \\
& 3 \tilde{\mu} y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} y_u^{pr} \overline{a_u^{pr}} + \overline{y_u^{pr}} (C_{\phi2\phi1} a_u^{pr} + 2 C_{\phi2^2} \tilde{\mu} y_u^{pr}) \right) \text{LF}_{5,1,-2} [m_u^r, m_q^p] + \\
& \frac{3}{4} g_1^2 y_e^{i1i2} \text{LF}_{2,1,-1} [\tilde{\mu}, m_1] - \frac{1}{2} g_1^2 y_e^{i1i2} \text{LF}_{3,1,-2} [\tilde{\mu}, m_1] + 2 C_{\phi2^2} g_1^2 y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] - \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{3,1,0} [\tilde{\mu}, m_1] - 4 C_{\phi2^2} g_1^2 y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] + \\
& \frac{3}{2} m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + 2 C_{\phi2^2} g_1^2 y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] - \\
& m_1 \tilde{\mu} g_1^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] + \frac{9}{4} g_2^2 y_e^{i1i2} \text{LF}_{2,1,-1} [\tilde{\mu}, m_2] - \\
& \frac{3}{2} g_2^2 y_e^{i1i2} \text{LF}_{3,1,-2} [\tilde{\mu}, m_2] + 6 C_{\phi2^2} g_2^2 y_e^{i1i2} \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] - \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{3,1,0} [\tilde{\mu}, m_2] - 12 C_{\phi2^2} g_2^2 y_e^{i1i2} \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + \\
& \frac{9}{2} m_2 \tilde{\mu} g_2^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + 6 C_{\phi2^2} g_2^2 y_e^{i1i2} \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] - \\
& 3 m_2 \tilde{\mu} g_2^2 y_e^{i1i2} \left(\overline{C_{\phi2\phi1}} + C_{\phi2\phi1} \right) \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] - \frac{1}{4} \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,-1} [\tilde{\mu}, m_e^r] - \\
& \frac{1}{2} \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} \text{LF}_{2,1,-1} [\tilde{\mu}, m_l^p] - m_1 g_1^2 a_e^{i1i2} \text{LF}_{1,1,1,0} [m_1, m_e^{i2}, m_l^{i1}] + \\
& \frac{1}{4} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{2,2,1,-1} [m_1, m_e^{i2}, m_l^{i1}] - g_1^2 y_e^{i1i2} \\
& \text{LF}_{1,1,1,-1} [m_1, m_e^{i2}, \tilde{\mu}] + \frac{1}{4} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{2,2,1,-1} [m_1, m_l^{i1}, m_e^{i2}] + \\
& \frac{1}{2} g_1^2 y_e^{i1i2} \text{LF}_{1,1,1,-1} [m_1, m_l^{i1}, \tilde{\mu}] - \frac{1}{2} C_{\phi2^2} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_e^{i2}] + \\
& \frac{1}{2} m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_e^{i2}] + \frac{1}{4} C_{\phi2^2} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_1, \tilde{\mu}, m_l^{i1}] - \\
& \frac{1}{4} m_1 \overline{C_{\phi2\phi1}} \tilde{\mu} g_1^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_l^{i1}] - \frac{3}{2} g_2^2 y_e^{i1i2} \text{LF}_{1,1,1,-1} [m_2, m_l^{i1}, \tilde{\mu}] - \\
& \frac{3}{4} C_{\phi2^2} g_2^2 y_e^{i1i2} \text{LF}_{2,2,1,-2} [m_2, \tilde{\mu}, m_l^{i1}] + \frac{3}{4} m_2 \overline{C_{\phi2\phi1}} \tilde{\mu} g_2^2 y_e^{i1i2} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_l^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{2,1,1,0} [m_e^{i2}, m_1, m_l^{i1}] + \\
& \frac{1}{2} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{3,1,1,-1} [m_e^{i2}, m_1, m_l^{i1}] - \\
& \frac{1}{2} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{2,1,1,0} [m_l^{i1}, m_1, m_e^{i2}] + \\
& \frac{1}{2} m_1 g_1^2 \left(C_{\phi2^2} a_e^{i1i2} + \overline{C_{\phi2\phi1}} \tilde{\mu} y_e^{i1i2} \right) \text{LF}_{3,1,1,-1} [m_l^{i1}, m_1, m_e^{i2}] \Big)
\end{aligned}$$